

Good, Better, Best

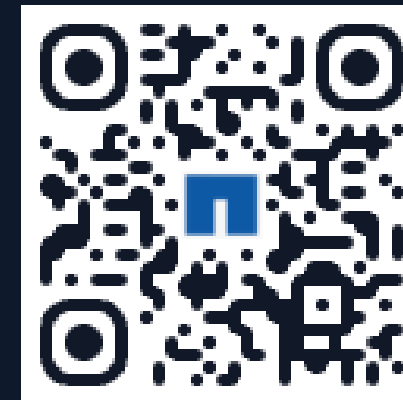
Choosing a RAG architecture with OpenSearch

Brian Graf · NetApp Instaclustr · OpenSearchCon NA



Resources and the Example Corp kit

Build and inspect each stage yourself.



Advanced RAG with OpenSearch

Resources covering model deployment, indexing, evidence packing, evaluation, hybrid retrieval, and conversation memory.

github.com/instaclustr/InstAcademy

The Example Corp kit

The corpus, the 300-query golden set, the loader, the scorer, and both chat forms. Standard library only — no pip installs, no framework.

For your own documents, adapt ingestion and build an evaluation set.

The repository includes the corpus, evaluation set, scripts, and setup instructions.
Recorded scores and model responses can vary with the query and configuration.

One assistant, three capabilities

GOOD

Standalone questions



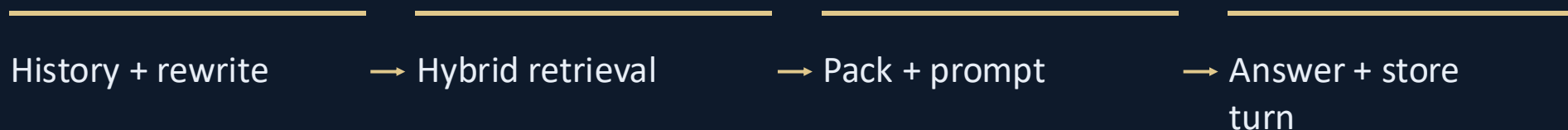
BETTER

Mixed terminology



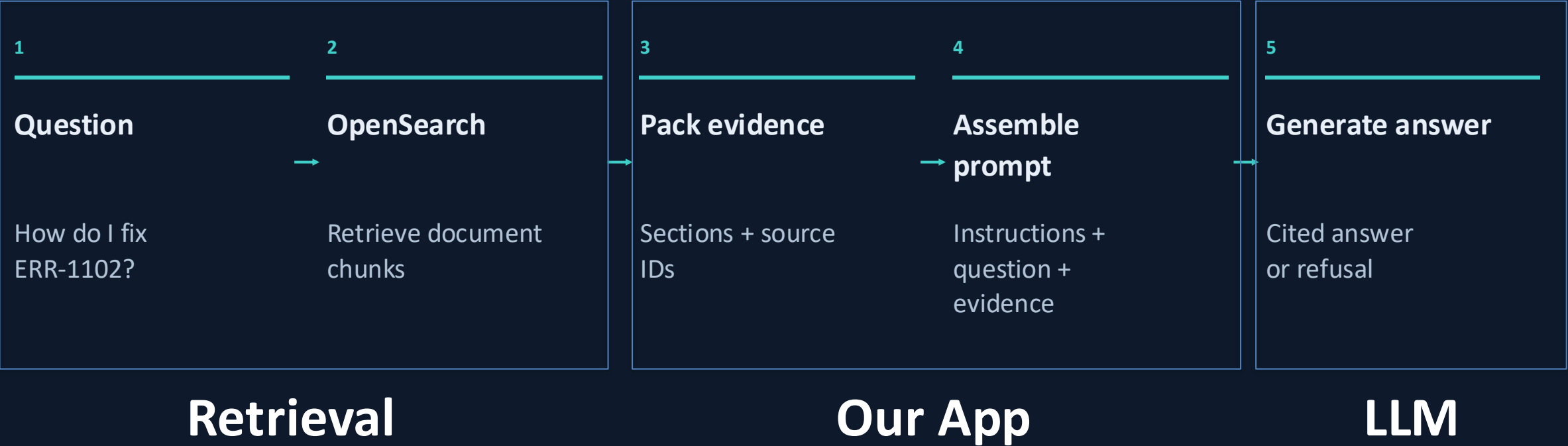
BEST

Contextual follow-ups



Each addition solves a different problem. Evaluate the capability your traffic needs.

Search finds evidence. The model writes



Under the hood: the RAG environment

The corpus and models behind the first RAG pipeline

OpenSearch 3.5.0

Index support-advrag-kb: 5,139 chunks from 1,079 source documents.

Example Corp corpus

A fictional support knowledge base: product docs, integration guides, API reference, 15 known issues.

Embedding model

all-MiniLM-L6-v2 in ML Commons. 384 dimensions, HNSW index, cosine similarity.

Embedding at ingest

The ingest pipeline embeds document chunks before indexing them.

Evidence packing

The application deduplicates parent sections and carries source IDs into the prompt.

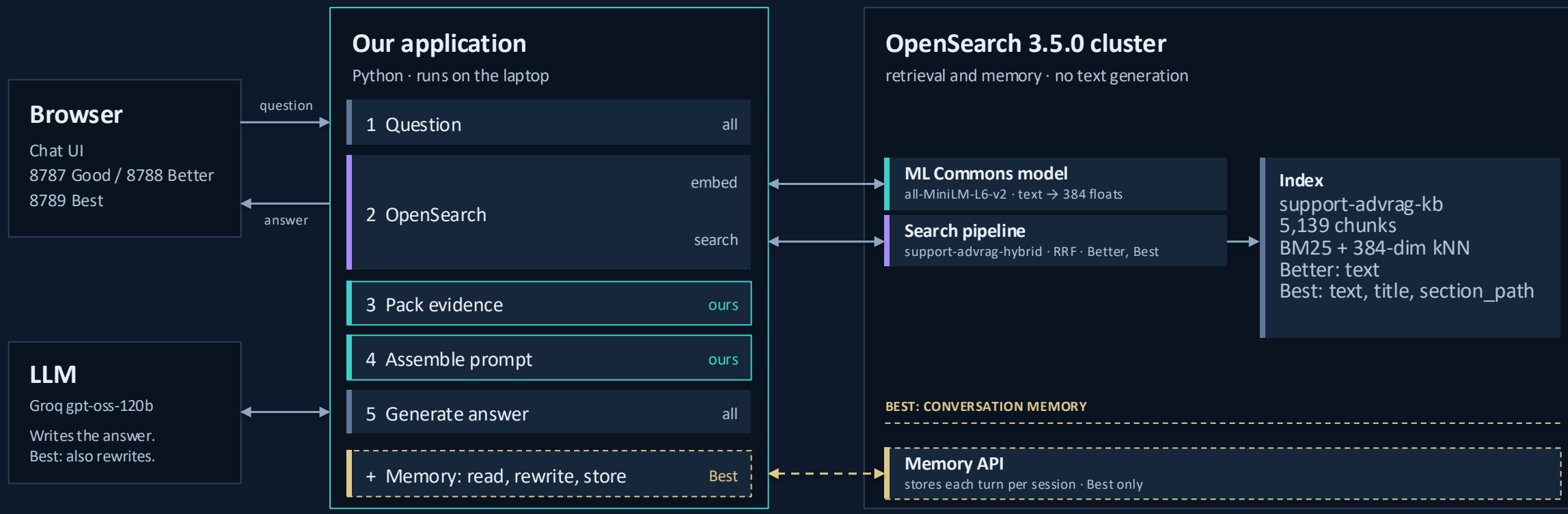
LLM

Groq gpt-oss-120b, temperature 0. Receives instructions, the question, and packed evidence.

Under the hood: how the pieces connect

Three systems, and our code in the middle

The application calls OpenSearch and Groq separately and moves the data between them.

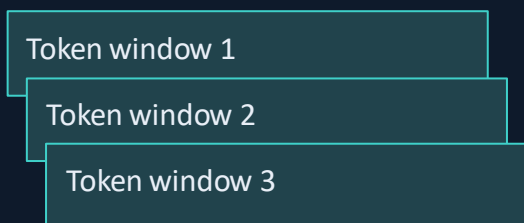


Good

Grounded answers with vector retrieval

Four ways to divide documents into chunks

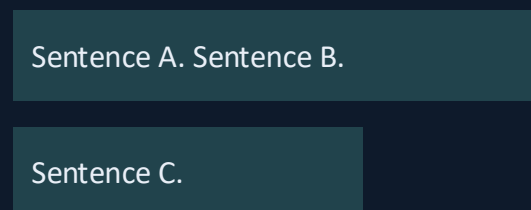
Fixed-size



Equal token windows,
often with overlap.

Predictable sizes; a boundary
can split an instruction.

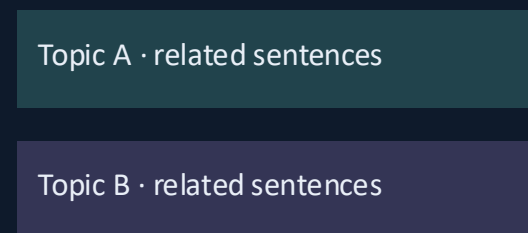
Sentence-based



Group whole sentences
within a budget.

Preserves sentences; headings
can separate from their
content.

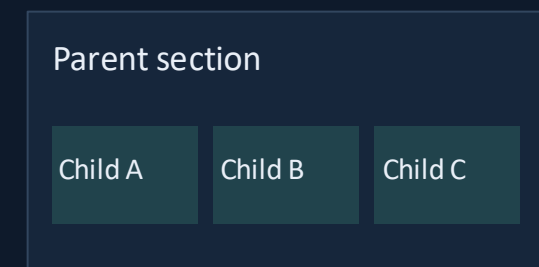
Semantic



Split where the meaning
changes.

Extra embedding work and
boundary thresholds to tune.

Parent-child



Search children; supply
the parent section.

We use this for
product documentation and
guides.

Choose boundaries for the document type. Parent-child also separates the search unit from the answer context.

A test set before a bigger architecture

1,079 documents

5,139 chunks

300 labeled questions

Synthetic Example Corp corpus. Resolved tickets provide the questions and labels; tickets stay out of the evaluation index.

RANK 1 A Not relevant	RANK 2 B ✓ Relevant	RANK 3 C Not relevant	RANK 4 D ✓ Relevant	RANK 5 E Not relevant
------------------------------------	----------------------------------	------------------------------------	----------------------------------	------------------------------------

HIT@5

1

At least one relevant result.

PRECISION@5

2 / 5 = 0.4

Two of five results are relevant.

RECIPROCAL RANK

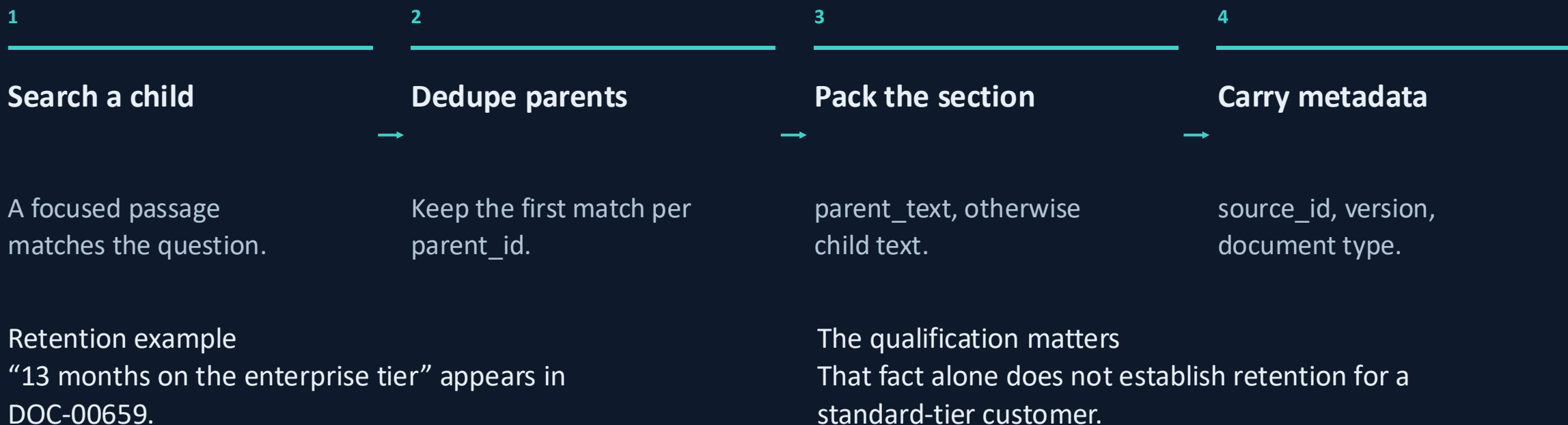
1 / 2 = 0.5

First relevant result is second.

Illustrative single query. Average these measures across the evaluation questions; mean reciprocal rank is MRR.

Measure retrieved evidence before judging the generated answer.

Good: small chunks, useful context



Known issues have no parent_text. The child-text fallback keeps them in the prompt.

Dev Tools: inspect retrieval before generation

1 · RUN IN OPENSEARCH DASHBOARDS → DEV TOOLS

Search the knowledge base

POST support-advrag-kb/_search

```
{
  "size": 3,
  "collapse": { "field": "parent_id" },
  "_source": ["source_id", "parent_id",
    "section_heading", "product_version",
    "text", "parent_text"],
  "query": {
    "neural": {
      "text_embedding": {
        "query_text": "how long are dashboard rendering audit events retained",
        "model_id": "YOUR_MODEL_ID",
        "k": 10
      }
    }
  }
}
```

2 · INSPECT THE RETURNED HIT

DOC-00659

source_id: DOC-00659
parent_id: DOC-00659#1
section_heading: Configuration
product_version: 5.0

Audit events ... retained for 13
months on the enterprise tier.

Live check: DOC-00659#1, version 5.0.
Dev Tools uses collapse here.
The browser deduplicates after search.

Five instruction blocks + evidence

BLOCK 1	SYSTEM
BLOCK 2	TOOLING (empty)
BLOCK 3	GOAL
BLOCK 4	RESTRICTIONS
BLOCK 5	ANSWER SCHEMA
RETRIEVED DATA	
PAYLOAD	EVIDENCE

Evidence from vector retrieval

Packed into Good's user message; source metadata condensed here

[API-010] POST /v2/alerts: ERR-4402, ERR-6601

[API-008] GET /v2/connectors: ERR-6601

[API-005] GET /v2/datasets: ERR-6601

For this short query, the evidence has no ERR-1102 fix.
Good should say it does not have that information.

Customer restrictions

Source	Version & Level Applicability	Documented limit
DOC-00095	4.8 / standard	25 dashboards
DOC-00330	4.8 / professional	50 dashboards
DOC-00847	5.0 / enterprise	100 dashboards

Customer: 4.8 / standard

Recorded answer: 25 dashboards on the standard tier [DOC-00095].

experiment: removing grounding

Grounding present

Grounding removed

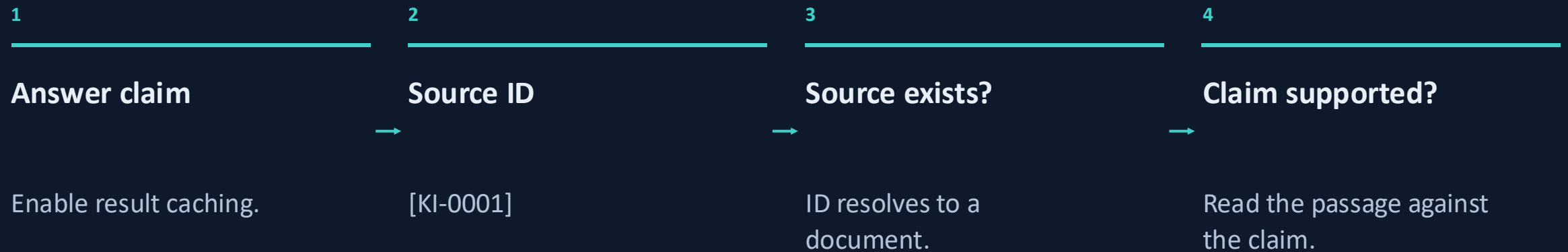
Recorded response: unsupported procedure

Open the Data Manager. Click Settings. Enable Result Caching. [KI-0001]

Confidence: medium

Citations and formatting can survive even when grounding fails.

A citation is a claim to check



“Confidence: medium” is model output. It is not a measured probability.

The two-message prompt

1 · OPENSEARCH RESPONSE

Retrieved fields

source_id
product_version
parent_text
or text

From the Dev Tools
response we just
inspected.

2 · APPLICATION ASSEMBLES

role: system

You are Example Corp's internal support assistant.

Answer from evidence; decline if the answer is missing.
Cite claims. End with Confidence: low, medium, or high.

role: user

Evidence:
Packed source sections, with IDs and metadata

[source_id] · document type · version · passage

Question: current input → answer with [source_id] citations

3 · ANSWER MODEL

Generate

Receives both
messages.

Returns an answer
with source
citations, or says the
information is
missing.

GOOD

Demo 1: grounded, stateless answers

Questions

- 01 How do I fix ERR-1102?
- 02 how long are dashboard rendering audit events retained
- 03 how do I reset my billing password?

Optional longer question adds “dashboard render timeout” and can succeed. Keep it separate from button 1.

Under the hood: the search request

“How do I fix ERR-1102?”

Good: vector search · one request, one retrieval leg, no search pipeline

POST /support-advrag-kb/_search

```
{
  "size": 8,
  "query": {
    "knn": {
      "text_embedding": {
        "vector": [-0.0890, -0.0171, ...], // 384 floats
        "k": 10
      }
    }
  },
  "_source": ["source_id", "parent_id", "title", "text",
    "parent_text", "doc_type", "product_version"]
}
```

Vector leg (kNN)

The question embedded as 384 floats. Returns the 8 nearest chunks by meaning.

Good: the short error-code query misses the fix

Question	Expected behavior	Observed September 24	Why
ERR-1102	Decline without supporting evidence	"I do not have that information." Confidence: low	KI-0001 absent from all 8 hits. API error lists contain no fix.
Audit retention	Keep the tier qualification	13 months on enterprise [DOC-00659]	The evidence supports enterprise. It does not establish standard-tier retention.
Billing password	Decline unsupported instructions	"I do not have that information."	Retrieved integration guides do not answer the requested task.

Better

Hybrid retrieval with BM25 and Semantic Search

Hybrid gives evidence two routes into the prompt

BM25 ranks

1. Known issue A

2. Guide B

3. Document C

Vector ranks

1. Document C

2. Known issue A

3. Document D

RRF result

A: $1/61 + 1/62 = 0.03252$

C: $1/63 + 1/61 = 0.03227$

B: $1/62 = 0.01613$

D: $1/63 = 0.01587$

RRF: $1 / (60 + \text{rank})$. Rank 1 $\rightarrow 1/61$; rank 2 $\rightarrow 1/62$; rank 3 $\rightarrow 1/63$. Missing $\rightarrow 0$.

RRF combines rank positions. Raw BM25 and vector scores never meet.

Hybrid request for ERR-1102

POST /support-advrag-kb/_search?search_pipeline=support-advrag-hybrid

```
{
  "size": 8,
  "query": {
    "hybrid": {
      "queries": [
        { "match": { "text": "How do I fix ERR-1102?" } },
        { "knn": { "text_embedding": {
          "vector": [-0.0890, -0.0171, ...], // 384 floats
          "k": 10 } } }
      ]
    }
  },
  "_source": ["source_id", "parent_id", "title", "text",
    "parent_text", "doc_type", "product_version"]
}
```

Hybrid results include the known issue

```
{
  "took": 34,
  "hits": {
    "total": {
      "value": 2683,
      "relation": "eq"
    },
    "max_score": 0.016393442,
    "hits": [
      {
        "_id": "KI-0001#0.0",
        "_score": 0.01639,
        "_source": {
          "source_id": "KI-0001",
          "parent_id": "KI-0001#0",
          "doc_type": "known-issue",
          "title": "ERR-1102: Dashboard render timeout",
          "text": "ERR-1102: Dashboard render timeout\n\nSymptom: Users encounter ERR-1102 (dashboard render timeout).\nRoot cause: Widget query exceeded the 60 second render budget\nWorkaround: Reduce widget count or enable result caching on the underlying dataset\nAffected versions: 4.8, 4.9\nFixed in: 5.0",
          "parent_text": null,
          "product_version": null
        }
      },
      {
        "_id": "API-010#2.0",
        "_score": 0.01639,
        "_source": {
          "source_id": "API-010",
          "parent_id": "API-010#2",
          "doc_type": "api-reference",
          "title": "POST /v2/alerts",
          "text": "POST /v2/alerts > Errors\n\n- ERR-4402\n- ERR-6601",
          "parent_text": "Errors\n\n- ERR-4402\n- ERR-6601",
          "product_version": null
        }
      }
    ]
  }
}
```

Rank	Score	source_id	Type	Title
1	0.0164	KI-0001	known-issue	ERR-1102: Dashboard render timeout
2	0.0164	API-010	api-reference	POST /v2/alerts
3	0.0161	API-008	api-reference	GET /v2/connectors
4	0.0161	DOC-00428	product-docs	How dashboard sharing works
5	0.0159	API-005	api-reference	GET /v2/datasets
6	0.0159	DOC-00429	product-docs	How PDF export works
7	0.0156	DOC-00430	product-docs	How mobile layouts works
8	0.0156	API-001	api-reference	GET /v2/dashboards

Packing the retrieved evidence

One block per parent_id. Use parent_text, with child text as the fallback.
Keep up to 3 sections within a 1,500-word budget. Preserve source IDs.

[KI-0001] ERR-1102: Dashboard render timeout

Root cause: Widget query exceeded the 60 second render budget.

Workaround: Reduce widget count or enable result caching
on the underlying dataset.

Affected versions: 4.8, 4.9. Fixed in: 5.0.

[API-010] POST /v2/alerts: ERR-4402, ERR-6601

[API-008] GET /v2/connectors: ERR-6601

The model request for this hybrid run

SYSTEM MESSAGE

You are Example Corp's internal support assistant.
Answer ONLY using the evidence provided below.
If the evidence does not contain the answer,
say you do not have that information.

After each claim, cite its `source_id`.
Be concise. End with Confidence: low,
medium, or high.

USER MESSAGE

Evidence: [KI-0001]
ERR-1102: Dashboard render timeout
Workaround: Reduce widget count or enable
result caching on the underlying dataset.
Affected: 4.8, 4.9. Fixed in: 5.0.

Also packed: API-010 and API-008 error lists.

Question: How do I fix ERR-1102?
Grounded answer with [source_id] citations:

Abbreviated actual messages. API blocks and some wording are condensed for readability.

The answer uses the known issue

Recorded answer:

To resolve ERR-1102 (dashboard render timeout),
reduce the load on the dashboard by either:

- Lowering the number of widgets displayed, or
- Enabling result caching on the underlying dataset.

If possible, upgrade to version 5.0 where the issue is fixed.
[KI-0001]

Confidence: high

The answer cites KI-0001. The other packed blocks
do not contain the requested fix.

ERR-6640: workaround, no fixed version

Question:

How do I fix ERR-6640 signed embed url expired?

Embed URLs expire after 10 minutes. The host page cached a URL longer than that. [KI-0013]

Generate embed URLs server side on every page load.
Never cache them. [DOC-00743] [DOC-00132]

Known-issue status

Affected versions: 4.9, 5.0, 5.1. Fixed in: not yet fixed. [KI-0013]

This is a workaround. The evidence names no version that permanently fixes the issue.

Production requirement: document permissions

1 · Identify the signed-in user

A production application must check which documents the signed-in user may read.

2 · KEYWORD SEARCH (BM25)

Search permitted documents

2 · VECTOR SEARCH

Search permitted documents

3 · Combine search results

→ Add sources to the prompt

→ Generate the answer

Documents the user cannot read

Excluded from both searches and the prompt.

BETTER

Hybrid improves coverage and precision in this run

Retriever	Hit@5	Precision@5	MRR
BM25	0.603	0.147	0.375
Vector	0.643	0.151	0.359
Hybrid	0.667	0.169	0.369

Recorded benchmark: 300 queries, k = 5, knowledge-base corpus. Results depend on the data and configuration.

Hybrid adds 2.4 percentage points of Hit@5 over vector in this recorded run.

Demo 2: exact codes and everyday language

QUESTIONS

- 01 How do I fix ERR-1102?
- 02 How do I fix ERR-6640 signed embed url expired?
- 03 my dashboards take forever to load
- 04 how do I reset my billing password?

Better: more evidence can still answer the wrong task

Question	Expected behavior	Observed September 24	Why
ERR-1102 (short query)	Find the known-issue fix	Fewer widgets or result caching. Fixed in 5.0. [KI-0001]	Hybrid returns KI-0001 first for the exact same input as Good.
ERR-6640	Known-issue workaround	Server-side URLs per page load; no client caching. [KI-0013]	Exact code helps retrieval. Vector also finds this issue.
Slow dashboards	Relevant troubleshooting	Timeout workaround and 5.0 fix [DOC-00337]	DOC-00142 was also packed. The answer cited DOC-00337.
Billing password	Still decline	Stripe credential-rotation instructions [GUIDE-022]	Failure: the cited instructions solve a different task.

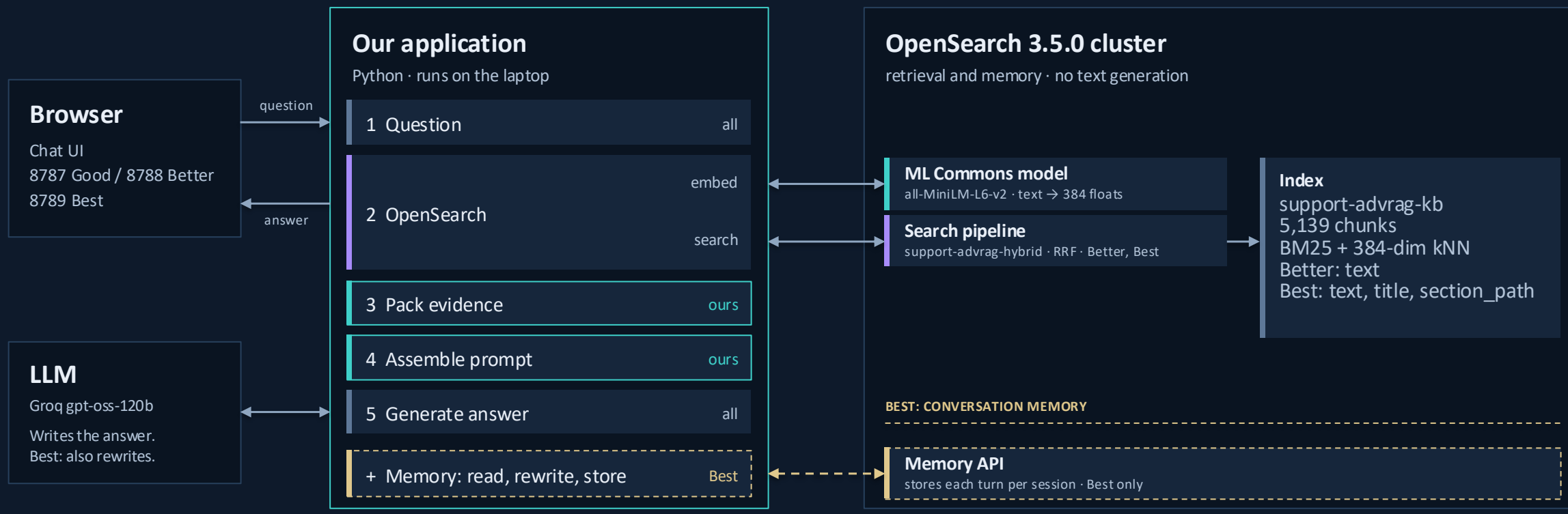
Best

Conversation memory and query rewriting

Under the hood: how the pieces connect

Three systems, and our code in the middle

The application calls OpenSearch and Groq separately and moves the data between them.



OpenSearch finds and remembers. The LLM writes.
Our code decides what the LLM gets to read.

OpenSearch memory: three API calls

Your application calls the ML Commons Memory API in OpenSearch.

WHEN THE APPLICATION CALLS		REQUEST TO OPENSEARCH		RETURNS
1 • Create a memory Once per conversation	→	POST <code>/_plugins/_ml/memory/</code> <code>{ "name": "Support conversation" }</code>	→	memory_id Keep this ID
2 • Read the history Before answering the next question	→	GET <code>/_plugins/_ml/memory/{memory_id}/messages</code>	→	messages[] Saved turns
3 • Save the turn After generating the answer	→	POST <code>/_plugins/_ml/memory/{memory_id}/messages</code> <code>{ "input": "...", "response": "..." }</code>	→	message_id Turn saved

Read history → Rewrite with history → Retrieve + answer → Save the turn

History restores the missing subject

1 · CURRENT MESSAGE

what about on version 5.0?

READ RECENT HISTORY

Earlier: How do I fix ERR-1102?

ML Commons conversation memory



2 · REWRITE WITH HISTORY

ERR-1102 fix version 5.0

3 · HYBRID RETRIEVAL

KI-0001: fixed in 5.0

PACK + PROMPT + GENERATE

Answer supported by the known issue.

↳ 4 · Store this input and response

Available as history on the next turn.

This app skips rewriting only when history is empty. With history, even complete questions are rewritten.

History supplies the subject. Retrieved evidence supplies the fix information.

“BEST” adds memory and customer context

Prompt structure

Best

MEMORY

Customer: 4.8, standard tier.

Recent turn: How do I fix ERR-1102?

GOAL: ERR-1102 fix version 5.0

EVIDENCE: [KI-0001] Fixed in: 5.0.

Best adds memory and customer context to the answer prompt.

Good / Better uses the simpler Evidence + Question layout.

Demo 3: a follow-up and a complete question

Demo Questions

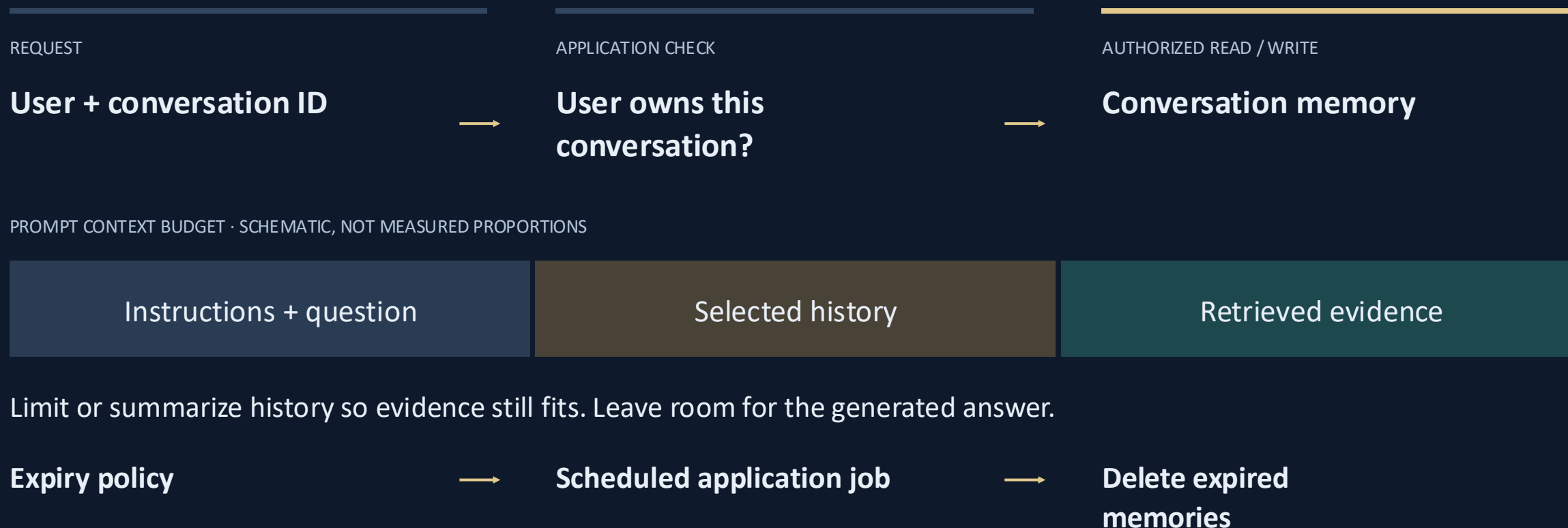
- 01 How do I fix ERR-1102?
- 02 what about on version 5.0?
- 03 my dashboards take forever to load
- 04 what is Example Corp's refund policy for annual contracts?

Best: useful follow-ups, but assumptions can travel

Turn	Expected behavior	Observed September 24	Why
1 · ERR-1102 (short query)	Grounded answer for version 4.8	Workaround for 4.8; fix in 5.0 [KI-0001]	Empty history: no rewrite call. The turn is saved afterward.
2 · Version 5.0?	Restore the issue before search	Rewrite: ERR-1102 fix version 5.0 Answer: fixed in 5.0. [KI-0001]	History supplies the error code. Evidence supplies the fix.
3 · Slow dashboards	Avoid unjustified carryover	Rewrite adds ERR-1102 and 5.0. Answer focuses on the 5.0 fix.	Prior context adds assumptions to a complete question.
4 · Annual refund policy	Decline unsupported policy details	"I do not have that information." Confidence: low	Memory does not supply a missing refund policy.

BEST

Conversation context has an operating cost



Conversation context adds state, token-budget decisions, and retention work.

Memory strategies for OpenSearch applications

Choose what history enters the prompt. Stored conversations can remain intact.

Window

Recent turns, word for word

Low overhead. Older details fall outside the prompt.

OpenSearch agents: SlidingWindowManager with max_messages.

Summary

A running summary of the conversation

Keeps the overall context. Compression can omit important details.

Update a summary with an LLM as the conversation grows.

Hybrid

Older history summarized, recent turns kept verbatim

Retains the long view and exact wording for follow-up questions.

OpenSearch agents: SummarizationManager with preserve_recent_messages.

Which version fits your users' questions?

Choose using examples of the questions people actually ask.

Good

A complete question in everyday language

“Why are my dashboards slow?”

Try vector search. Check that the answer uses relevant sources.

Better

Exact codes, product names, or varied wording

“How do I fix ERR-6640?”

Test hybrid search. Check for matches that solve the wrong problem.

Best

A follow-up that depends on earlier messages

“What about on version 5.0?”

Add conversation memory. Check that the rewrite preserves the user's intent.

Every version needs document permissions and clear prompt instructions.

Search operations, LLM calls, and ongoing work

Hybrid combines vector search and BM25, then merges their ranked results.

	SEARCH OPERATIONS	LLM CALLS PER QUESTION	EXTRA TO MANAGE
Good	Vector search Matches meaning	1 call Write the answer	No saved conversation
Better	Vector search + BM25 Meaning + words/codes Fusion combines the ranked lists	1 call Write the answer	Hybrid search pipeline settings
Best	Hybrid + broader lexical fields text, title, section_path Rewrite when history exists	1 or 2 calls 1 answer call + 1 rewrite with history	Pipeline + history User ownership History limits / expiry

All three need indexed documents, an embedding model, and an answer model.

Counts are for this application. Embedding-model calls are separate from LLM calls.

Good, Better, Best: the takeaways

Each stage addresses a different failure in the answer pipeline.

Good

Good grounds answers in retrieved evidence

Vector search matches meaning. Short error codes can miss the fix.

When the evidence is insufficient, the answer should say so.

Better

Better combines meaning with exact terms

Hybrid search finds the ERR-1102 fix for the same short query.

A relevant-looking source can still answer the wrong task.

Best

Best carries context into follow-up questions

History restores the subject before retrieval.

This demo also adds customer context and broader lexical fields.

Evaluate the evidence and the answer with the questions your users ask.

Deck and links to do this yourself:



Thank you

Brian Graf

NetApp Instaclustr

CONNECT

[LinkedIn: thebriangraf](#)

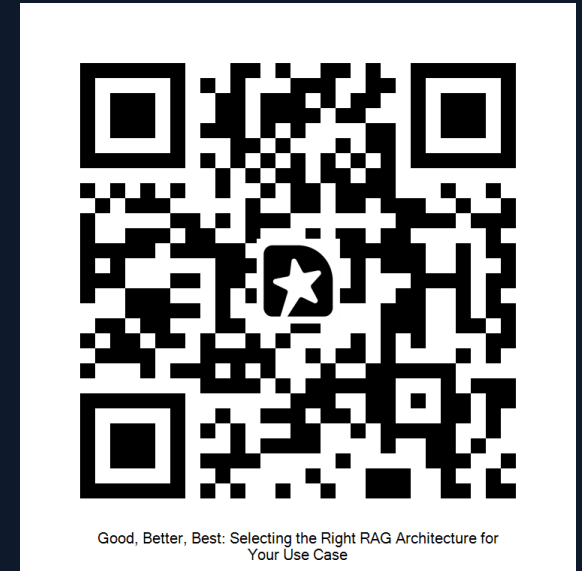
[GitHub : @TheBrianGraf](#)

[X : @thebriangraf](#)

RESOURCES

[InstAcademy ntap.fyi/courses](#)

[Docs docs.opensearch.org](#)



Questions?